Mobile WiMAX: Meet 802.16e

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The world was revolutionised with the advent of mobile phones. Along came the concepts of text messaging and missed calls. These ensured we always stayed connected to our contacts.

We are about to experience yet another revolution. Worldwide interoperability for microwave access (WiMAX) has the potential to do to the internet exactly what mobiles did to telephones - to make it wireless on the large scale. Additionally, it can also be used to help mobile telecommunications.

What was wrong with WiFi?

A popular misconception is that WiMAX is aimed at replacing WiFi. This isn't entirely true. WiFi is only concerned with creating a wireless pocket of connectivity around the router, instead of being strangled with a wire.

WiMAX, on the other hand, takes wireless to the maximum (although that isn't what it stands for), by facilitating wireless broadband connectivity of up to 3 Mbps spread over large areas. WiFi can be used to create wireless local area networks and provide internet connectivity to computers in the range of a router however it doesn't provide a viable means to connect computers over a wide range, say that of a city. WiFi can thus be thought of as a wireless alternative to Ethernet networks, while WiMAX is an alternative to technologies such as DSL.

WiMAX

WiMAX is an acronym for Worldwide Interoperability for Microwave Access. It is a relatively new technology, having been first introduced in 2004 as 802.16, more popularly known now as 'fixed WiMAX'. It was created as a technology to provide the last-mile connectivity for broadband access. This means that it can potentially be used as a replacement for current broadband network systems such as DSL, landline or cable connectivity.

Where WiMAX wins out here is due to the fact that it is wireless, and as such huge savings can be expected from implementing this is place of wired technologies. Just like cellular-phone towers, setting up one WiMAX tower has the potential to cover a large area, and as such, can also be used to provide connectivity to some of the more remote areas where wired connectivity is currently unavailable, infeasible or expensive.

In a revision accepted in 2005 named 802.16e, many improvements were made to the standard, some of which were to enable the use of WiMAX in mobile installations.

IEEE 802.16e-2005

This is indeed the full standard name for "Mobile WiMAX". This amendment of the WiMAX standard is perhaps most popular for the addition of features enabling its use in mobile installations. The standard describes itself as "Physical and Medium Access Control Layers for Combined Fixed and Mobile Operation in Licensed Bands", changing the title of the 802.16 standard to "Air Interface for Fixed and Mobile Broadband Wireless Access Systems". WiMAX, by itself, promises wonders, then why all the hype about this new amendment?

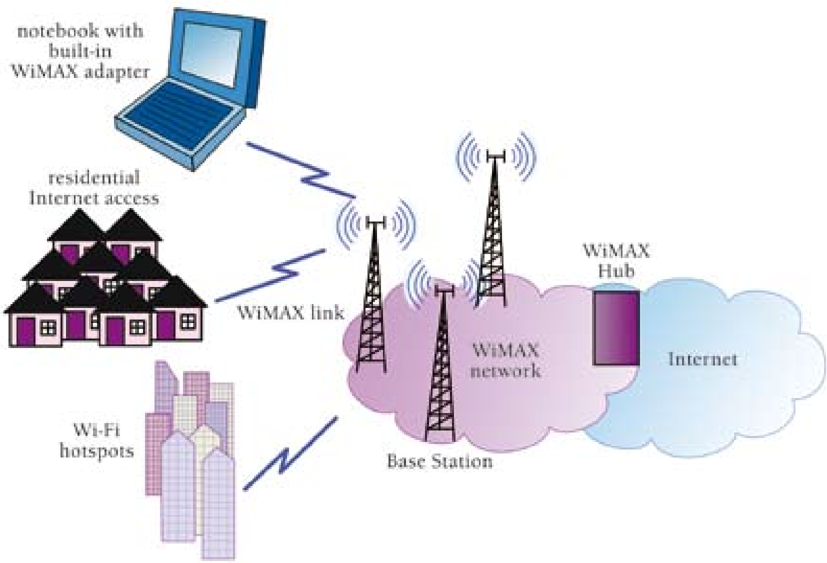
Consider this – you have full speed broadband access on the go; surfing wirelessly at home, you could check your email on the way to work, and continue working once there without ever needing to connect a wire! What if this connectivity existed everywhere, you would never have to worry about searching for hotspots. The whole city would be a hotspot. But then, since this is India, you'll have to imagine waiting for another three decades before this becomes a reality.

One of the key technologies for facilitating mobile access of WiMAX networks is a scheme called SOFDMA (Scalable Orthogonal Frequency Division Multiple Access) which is a scalable form of the system used in 802.16-2004 OFDMA (Orthogonal Frequency Division Multiple Access).

SOFDMA Deconstructed

Scalable Orthogonal Frequency Division Multiple Access, quite a mouthful! However if we break it down into its components it begins to sound much more sensible.

Scalable: Scalability is important when considering any technology. Especially for WiMAX which can be used to provide connectivity for as far as 50 miles, and may



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